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Development and validation of an interpretable machine learning model for postoperative radiotherapy decision-making in ypN0 breast cancer after neoadjuvant chemotherapy: a real-world study

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Abstract

Objective The subset of breast cancer patients with pathologic node-negative status (ypN0) after neoadjuvant chemotherapy (NAC) who benefit from postoperative radiotherapy (PORT) through a reduced risk of recurrence remains poorly defined. This study aimed to develop and validate an interpretable machine learning (ML) model to perform risk stratification for recurrence among ypN0 patients who all received PORT.

Methods We conducted a retrospective analysis of 1450 breast cancer patients treated between January 2017 and January 2024. All patients received NAC, underwent radical surgery confirming ypN0 status, and subsequently received PORT. Based on follow-up outcomes after PORT, patients were classified as 'recurrence' or 'no-recurrence'. From 20 initial clinicopathological variables, feature selection was performed using LASSO regression, stepwise logistic regression, and the Boruta algorithm, retaining 16 key features identified by at least two methods. Predictive models were constructed using ten machine learning algorithms, with hyperparameters optimized via particle swarm optimization. Model performance was assessed using the area under the receiver operating characteristic curve (AUC), calibration curves, decision curve analysis (DCA), and standard metrics including accuracy, sensitivity, specificity, and F1-score. The optimal model was interpreted using the SHapley Additive exPlanations (SHAP) framework.

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Results The Gradient Boosting Machine (GBM) model demonstrated superior predictive performance, achieving an AUC of 0.945 (95% CI: 0.917–0.972) on the test set. SHAP analysis identified breast reconstruction, perineural invasion, and the chemotherapy-to-surgery interval as the three most influential predictors. These features primarily exerted independent effects on recurrence risk.

Conclusion We developed and validated a highly accurate and interpretable ML model to stratify recurrence risk in ypN0 breast cancer patients after PORT. Functioning as a risk-stratification tool within this uniformly treated cohort, it may help identify patients at high risk of recurrence despite receiving PORT, who could be candidates for intensified surveillance or consideration of adjuvant therapy escalation. Its role in personalizing radiotherapy decisions requires prospective validation in studies including untreated control groups.

Keywords Breast cancer, Neoadjuvant chemotherapy, ypN0 status, Postoperative radiotherapy, Machine learning, SHAP

Introduction

Breast cancer, the most frequently diagnosed malignancy in women globally [1], exhibits considerable heterogeneity, necessitating increasingly personalized and precise treatment strategies [2, 3]. Neoadjuvant chemotherapy (NAC) is a standard approach for locally advanced breast cancer, primarily aiming to achieve a pathological complete response (pCR) [4]. Conversion to axillary node-negative status (ypN0) following NAC is a key indicator of treatment efficacy and is strongly correlated with improved long-term survival. Contemporary evidence, including a study reported at the 2024 San Antonio Breast Cancer Symposium (SABCS), suggests that for patients with 1–3 residual positive lymph nodes after NAC, postmastectomy radiation therapy does not significantly improve overall survival or provide a clinically meaningful reduction in locoregional recurrence within modern multimodal treatment paradigms [5]. In contrast, the clinical utility of regional nodal irradiation for patients who achieve ypN0 status post-NAC remains a subject of significant debate, posing a critical challenge in adjuvant treatment decision-making.

Current international guidelines present conflicting recommendations regarding PORT for ypN0 patients, largely relying on low-level evidence from retrospective analyses [6–8]. Some guidelines suggest considering PORT for ypN0 patients initially staged as clinical III or with lymphovascular invasion (LVI). However, the predictive utility of traditional clinicopathological factors (e.g., initial tumor size, LVI status) underlying these recommendations is limited and inconsistent across studies [9, 10]. More critically, existing research predominantly addresses the population-level question of “whether to irradiate,” [11] failing to resolve the central clinical dilemma: among ypN0 patients uniformly scheduled for PORT, how can we accurately stratify risk and identify those individuals who remain at high risk of recurrence despite receiving PORT [12]? The absence of a robust predictive tool leads to a dual clinical problem: patients unlikely to benefit are exposed to unnecessary

treatment-related toxicities (e.g. skin reactions, cardiovascular risks) and impaired quality of life, while truly high-risk patients may be undertreated, resulting in sub-optimal resource allocation.

Machine learning (ML), a cornerstone of artificial intelligence, holds great promise for medical prognostic modeling due to its capacity to manage high-dimensional data and identify complex, non-linear relationships [13–15]. However, within this field, most current ML applications focus on predicting pCR or general recurrence risk [16]. To our knowledge, no interpretable ML model has yet been developed and validated to address the precise clinical question of stratifying ypN0 patients by their likelihood of benefit from radiotherapy. Furthermore, the majority of existing predictive models operate as “black boxes,” with opaque decision-making processes that considerably hinder clinical trust and adoption [17, 18].

Therefore, this study aims to address this notable gap. Utilizing large-scale real-world data from ypN0 patients who all received PORT, we developed and validated the first interpretable machine learning model specifically designed not to decide “if” but to stratify “who” is at high risk of recurrence within this uniformly treated population. By systematically evaluating ten machine learning algorithms and employing explainable AI (XAI) techniques such as SHAP, this research prioritizes not only predictive accuracy but also transparency in elucidating key predictive features and their functional relationships with the outcome. We anticipate that this model will serve as an effective clinical decision-support tool, providing insights for risk-adapted management within this specific patient cohort, ultimately promoting precision medicine for ypN0 breast cancer patients and avoiding both overtreatment and undertreatment.

Materials and methods

Study population and data source

Data were retrospectively collected for breast cancer patients who received NAC followed by radical mastectomy at the Department of Thyroid, Breast, and Vascular

Surgery, First Affiliated Hospital of Air Force Medical University, between January 2017 and January 2024. The study cohort was restricted to patients with pathologically confirmed ypN0 status after surgery who subsequently received adjuvant PORT. NAC and radiotherapy regimens adhered strictly to contemporary NCCN guidelines. Radical mastectomy procedures followed standard surgical protocols. All data were extracted from the hospital's Electronic Medical Record (EMR) system. This cohort observational study used de-identified retrospective data and posed minimal risk to patients; therefore, the ethics committee waived informed consent. It is important to note that as all included patients received PORT, this study is designed to develop a prognostic risk-stratification model within this uniformly managed population, rather than a predictive model of treatment effect in the absence of a control group. Recurrence was defined as the first documented occurrence of locoregional recurrence (ipsilateral breast/chest wall or regional lymph nodes) or distant metastasis after completion of PORT. The study protocol complied with the ethical principles of the Declaration of Helsinki (1995) and received approval from the Institutional Review Board of the First Affiliated Hospital of Air Force Medical University (Approval No.: KY20232266-F-1).

Inclusion and exclusion criteria

Inclusion criteria were: (1) Histopathological confirmation of invasive breast carcinoma; (2) Completion of all planned NAC cycles followed by radical surgery; (3) Postoperative pathological assessment confirming ypN0 status; (4) Administration of standard adjuvant PORT following radical surgery; (5) Availability of complete clinical records detailing NAC and postoperative pathology.

Exclusion criteria encompassed: (1) History of other primary malignancies (non-breast origin) or presence of distant metastases at diagnosis; (2) Treatment with palliative intent; (3) Postoperative follow-up duration of less than 6 months or loss to follow-up; (4) Significant missing data (> 20%) for key variables or records containing obvious logical inconsistencies. The flowchart of patient selection and model construction is illustrated in Fig. 1.

Variables and data preprocessing

Twenty variables (16 categorical, 4 continuous) were initially considered. Variables with missing values (all missingness < 20%) were imputed using the Random Forest algorithm, resulting in a complete dataset. Random Forest imputation was chosen for its ability to handle non-linear relationships and mixed data types, providing a more accurate estimate than simple imputation methods for complex clinical data. To mitigate multicollinearity, variance inflation factor (VIF) analysis was performed,

and variables with $VIF > 5$ were excluded. Subsequent feature selection employed three distinct methods: Least Absolute Shrinkage and Selection Operator (LASSO) regression, stepwise logistic regression, and the Boruta algorithm. For LASSO, the optimal regularization parameter (λ) was determined via 10-fold cross-validation, using the plateau of the AUC curve as the stopping criterion. To enhance model robustness, only features identified as significant by at least two of the three selection methods were retained for final model building. Categorical variables were encoded as dummy variables prior to modeling.

Note on potential bias: This feature selection process was applied to the entire dataset prior to splitting into training and test sets. While this is a common approach in retrospective modeling, it may introduce optimistic bias in subsequent performance estimates, as information from the test set could indirectly influence feature selection. The reported model performance should therefore be interpreted with this limitation in mind.

Model development and validation

The cohort of 1450 patients was randomly partitioned into a training set (70%) and an internal test set (30%) to evaluate model generalizability. Stratified sampling was used during this partition to preserve the proportion of recurrence cases in both the training and test sets, addressing potential class imbalance. Model hyperparameters were tuned using grid search with 10-fold cross-validation. Ten machine learning algorithms were implemented: Logistic Regression, Support Vector Machine (SVM), Gradient Boosting Machine (GBM), Neural Network, Random Forest (RF), eXtreme Gradient Boosting (XGBoost), K-Nearest Neighbors (KNN), Adaptive Boosting (AdaBoost), Light Gradient Boosting Machine (LightGBM), and CatBoost. Hyperparameter optimization was conducted using the Particle Swarm Optimization (PSO) algorithm. For the GBM model, key hyperparameter ranges explored by PSO included: number of trees (100–1000), interaction depth (3–10), shrinkage (0.01–0.1), and minimum observations in a node (5–20). PSO was chosen for its efficiency in navigating high-dimensional parameter spaces and its global search capability, which helps avoid local optima—a common challenge when tuning complex ensemble models like GBM. Early stopping was incorporated to prevent overfitting. Model performance was rigorously evaluated based on the area under the receiver operating characteristic curve (AUC), calibration curves, decision curve analysis (DCA), accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and F1-score.

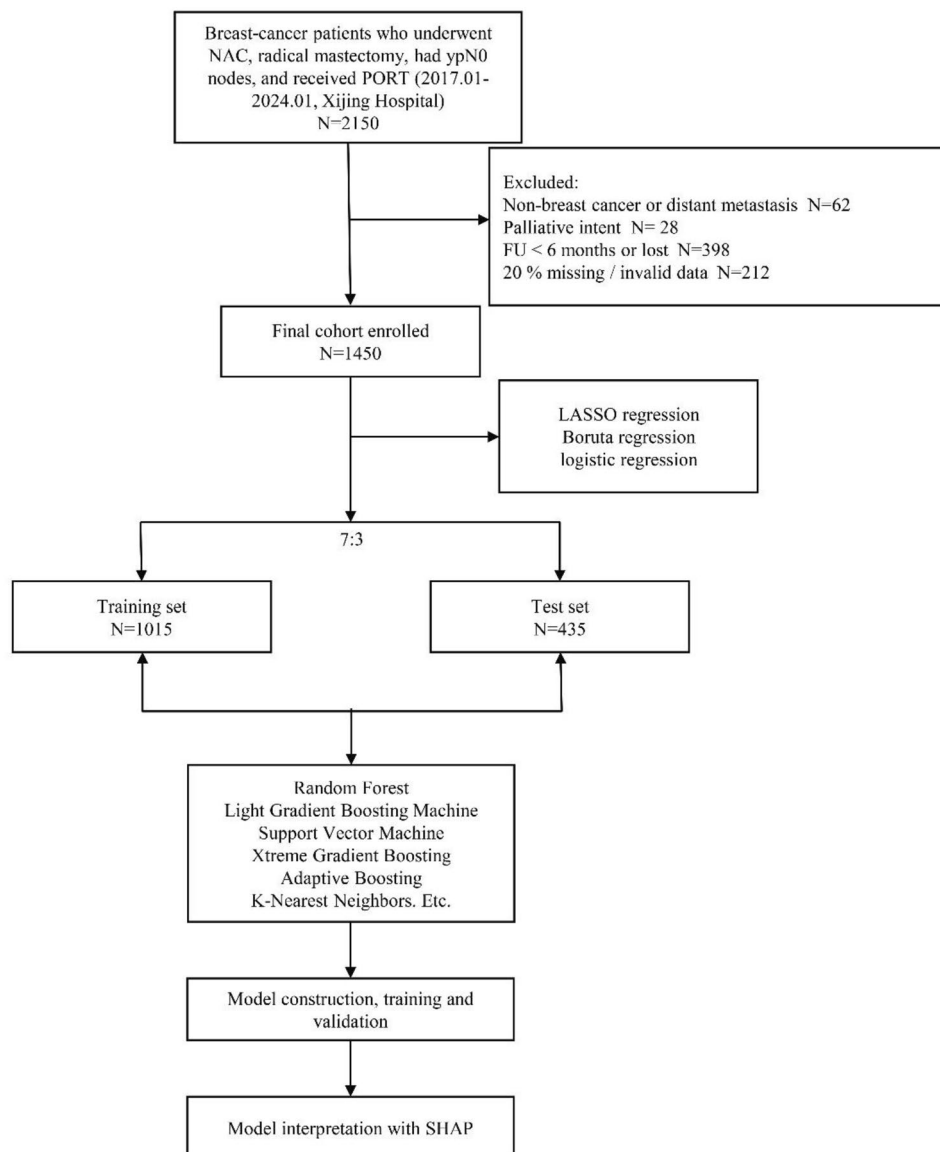


Fig. 1 Flowchart of our study procedure

Model interpretation

The SHapley Additive exPlanations (SHAP) method was applied to interpret the predictions of the final model [19]. This approach has been effectively utilized in clinical tabular data analysis within streamlined machine learning frameworks such as PyCaret [20], demonstrating its utility in enhancing model interpretability in healthcare contexts. Rooted in cooperative game theory, SHAP quantifies the marginal contribution of each feature to the individual prediction, providing both global insight into overall feature importance and local explanations for specific predictions, thereby clarifying the direction and magnitude of each variable's influence.

Statistical analysis

All analyses were performed using R software (version 4.4.1). Specific packages included: *glmnet* (LASSO), *Boruta* (Boruta algorithm), *e1071* (SVM), *gbm* (GBM), *caret*, *XGBoost*, *nnet* (Neural Network), *adabag* (AdaBoost), *lightgbm*, *CatBoost*, *pso* (Particle Swarm Optimization), *rmda* (DCA), *riskRegression* (calibration curves), and *kernelshap* (SHAP analysis). Continuous variables are summarized as mean $\pm$ standard error, and categorical variables as frequencies (percentages). Group comparisons for baseline characteristics utilized the Chi-squared test, with a two-sided P-value < 0.05 considered statistically significant.

Results

Patient characteristics

The study included 1450 breast cancer patients treated with NAC and radical surgery between January 2017 and January 2024, who achieved ypN0 status and received adjuvant PORT. The overall recurrence rate in the cohort was 36.0% (522/1450). For model development, the primary endpoint was recurrence after PORT. For analytical purposes, and acknowledging this is an observational association, patients without recurrence were classified as a ‘low-risk’ group ($n=928$), while those with recurrence were classified as a ‘high-risk’ group ($n=522$). It is acknowledged that this is a surrogate endpoint based on observational association. Baseline characteristics are detailed in Table 1. Patients in the high-risk group were younger (median age 40.37 years vs. 49 years) and exhibited significant differences ($P<0.05$) in molecular subtype, tumor size, perineural invasion status, and other features compared to the low-risk group, indicating underlying heterogeneity in tumor biology and clinical presentation.

Model performance and selection

Feature selection integrating Boruta, LASSO regression, and stepwise logistic regression identified 16 robust predictors consistently significant across at least two methods. These features were used to train the ten machine learning models. Performance metrics on the held-out test set are presented in Table 2. Most models demonstrated strong performance across multiple evaluation criteria. Considering the clinical imperative to minimize false positives (avoiding overtreatment in patients who would not recur), models with high specificity were prioritized. The F1-score, balancing precision and recall, was also considered crucial given the class distribution. The Gradient Boosting Machine (GBM) model showed the most favorable performance on the internal test set, with an AUC of 0.945 (95% CI: 0.917–0.972), significantly outperforming other algorithms (Fig. 2). Although CatBoost demonstrated a comparable AUC (0.946), the GBM model was selected as optimal due to its superior specificity (0.950) and negative predictive value (0.949), which are clinically prioritized to minimize false positives (i.e., avoiding overtreatment in patients unlikely to recur). This balance of metrics, alongside slightly better interpretability within the SHAP framework, rendered GBM more suitable for potential clinical translation.

Assessment of overfitting and generalizability

The high performance on the independent test set suggests effective generalization. We further addressed overfitting concerns by noting the close agreement between the model’s performance on the training set (AUC: 0.958) and the test set (AUC: 0.945). Additionally, the

Table 1 Comparison of baseline characteristics between patients with and without recurrence

Variable	Control N=928	Recurrence N=522	p-value
Breast, n(%)			0.046 ³
Left-sided	479(52)	302(58)	
Right-sided	443(48)	215(41)	
Bilateral	6(0.6)	5(1.0)	
ER, n(%)			0.001 ²
negative	503(54)	237(45)	
positive	425(46)	285(55)	
PR, n(%)			<0.001 ²
negative	548(59)	198(38)	
positive	380(41)	324(62)	
Her2, n(%)			<0.001 ²
0	153(16)	89(17)	
1+	109(12)	46(8.8)	
2+	140(15)	172(33)	
3+	526(57)	215(41)	
WHO, n(%)			<0.001 ³
0	30(3.2)	0(0)	
1	13(1.4)	0(0)	
2	404(44)	310(59)	
3	481(52)	212(41)	
Nur, n(%)			<0.001 ²
n	754(81)	522(100)	
y	174(19)	0(0)	
Vessel, n(%)			0.96 ²
n	637(69)	359(69)	
y	291(31)	163(31)	
Ps2, n(%)			0.52 ²
n	648(70)	356(68)	
y	280(30)	166(32)	
Endo, n(%)			<0.001 ²
n	610(66)	240(46)	
y	318(34)	282(54)	
Target, n(%)			<0.001 ²
n	251(27)	303(58)	
y	677(73)	219(42)	
Age, Median(IQR)	49.000(39.500–55.500)	40.370(35.393–45.182)	<0.001 ⁴
Rtime, Median(IQR)	34.000(33.000–39.000)	38.098(36.000–41.039)	<0.001 ⁴

Table 1 (continued)

Variable	Control N= 928	Recurrence N= 522	p-value
Other, Median(IQR)	0.400(0.230–0.600)	0.263(0.150–0.450)	< 0.001 ⁴
interval, Median(IQR)	2.000(1.000–3.000)	2.000(1.292–2.772)	0.034 ⁴

Variable definitions in pathological and treatment characteristics

WHO grade: WHO histopathological grading system

Nur: Perineural invasion status; “yes” indicates presence of tumor cells within the perineural space

Vessel: Lympho-vascular invasion status; “yes” denotes unequivocal tumor emboli within endothelial-lined spaces

Ps2: Estrogen-regulated protein 2 expression; routinely assessed by immunohistochemistry and reported as positive or negative

Endo: Adjuvant endocrine therapy; “yes” if any hormonal agent (e.g., tamoxifen, aromatase inhibitor) was administered

Target: Targeted therapy; “yes” if biological agents directed against specific molecular targets (e.g., HER2-directed therapy) were given

Rtime: Follow-up time (months) from the date of primary surgery to the last clinical visit or death

Other: Aggregate percentage of additional recognized risk factors not itemized above

Interval: Time interval (weeks) between completion of neoadjuvant chemotherapy and definitive surgical resection

calibration curve (Fig. 2C) demonstrates good agreement between predicted probabilities and observed event rates across risk deciles, indicating the model provides well-calibrated risk estimates rather than merely memorizing the training data.

Model interpretability

Despite the power of ML, its perceived “black-box” nature can limit clinical adoption [21]. SHAP analysis was employed to enhance the interpretability of the optimal GBM model, identifying key drivers of predictions and quantifying their contributions [22]. SHAP values indicate the magnitude and direction (positive or negative) of each feature’s impact on the predicted outcome (recurrence risk). Analysis revealed breast reconstruction, perineural invasion, and the chemotherapy-to-surgery interval as the top three contributors to the model’s predictions (Fig. 3). The strong association with breast reconstruction warrants attention to potential confounding by patient factors like general health status. Partial Dependence Plots suggested that these key features act largely independently. Furthermore, local explanation plots for representative cases (one recurrent, one non-recurrent) illustrated how specific variable values shaped individual predictions (Fig. 4), enhancing clinical applicability. Follow-up time was not included as a modeling variable for two primary reasons: firstly, its subjectivity and variability due to clinical scheduling and patient compliance; secondly, as a static classification model, incorporating time-to-event data without special handling (e.g., as a time-dependent covariate) could

introduce bias. Future studies employing survival analysis frameworks may address this. In summary, SHAP provided transparent, quantitative insights into the model’s decision-making process, identifying key risk factors and suggesting independent mechanisms, thereby facilitating trust and potential clinical translation.

Discussion

This real-world study developed and validated the first interpretable machine learning model specifically designed to stratify recurrence risk among ypN0 breast cancer patients who all received postoperative radiotherapy (PORT) following NAC. The GBM model demonstrated strong internal predictive performance (AUC: 0.945). Therefore, it should be viewed strictly as a risk-stratification tool within this uniformly managed ypN0 population that all received PORT. It does not predict the benefit of radiotherapy itself, but identifies a subset of patients who remain at high risk despite standard adjuvant treatment. Crucially, by leveraging the SHAP framework, we achieved high model interpretability, delineating key predictors, their effect sizes, and directional influences, thereby providing a transparent evidence base to inform individualised clinical decisions.

Therefore, our model is most accurately positioned as a highly accurate, risk-stratification tool within a uniformly managed ypN0 population. It identifies a subset of patients who, despite receiving standard adjuvant PORT, remain at significantly high risk of recurrence. This identification is of paramount clinical value in itself, as it pinpoints candidates who may warrant treatment intensification or more vigilant follow-up. We acknowledge that the most robust demonstration of predictive value for PORT benefit would require a control group without PORT to assess for a treatment-by-biomarker interaction. Our model provides a validated candidate tool for prospectively identifying such patients in future randomized trials.

Methodologically, the superior performance of our GBM model over traditional statistical approaches aligns with the growing application of ML in medical prediction [23]. The key innovation of our work lies in reframing the prediction target from the generic “risk of recurrence” to the more specific and clinically pertinent task of “recurrence risk stratification within a uniformly managed cohort that all received PORT.” While previous studies, such as that by Javanmard Z et al. using ML for survival prediction, involved heterogeneous populations [24], our model precisely targets the clinically ambiguous ypN0 group and incorporates comprehensive perioperative metrics. Notably, existing genomic assays (e.g., Oncotype DX®, MammaPrint®), while valuable for prognosticating overall recurrence risk, are not designed nor validated for predicting response to radiotherapy and entail substantial

Table 2 Evaluation indicators in testing set of 10ML models

Model	Accuracy	Sensitivity	Specificity	Precision	F1	PPV	NPV	AUC(95%CI)
Logistic	0.853	0.859	0.849	0.761	0.807	0.859	0.849	0.881(0.847–0.915)
SVM	0.931	0.929	0.932	0.884	0.906	0.929	0.932	0.943(0.915–0.972)
GBM	0.935	0.91	0.95	0.91	0.91	0.986	0.949	0.945(0.917–0.972)
NeuralNetwork	0.848	0.904	0.817	0.734	0.81	0.904	0.817	0.904(0.872–0.935)
RandomForest	0.942	0.936	0.946	0.907	0.921	0.936	0.946	0.941(0.915–0.966)
Xgboost	0.876	0.897	0.863	0.787	0.838	0.897	0.863	0.934(0.907–0.960)
KNN	0.917	0.846	0.957	0.917	0.88	0.846	0.957	0.901(0.871–0.932)
Adaboost	0.857	0.821	0.878	0.79	0.805	0.985	0.878	0.908(0.878–0.937)
LightGBM	0.91	0.917	0.906	0.846	0.88	0.917	0.906	0.938(0.912–0.964)
CatBoost	0.947	0.929	0.957	0.924	0.927	0.986	0.957	0.946(0.919–0.973)

Note: Values for Sensitivity, Specificity, etc., are point estimates on the test set

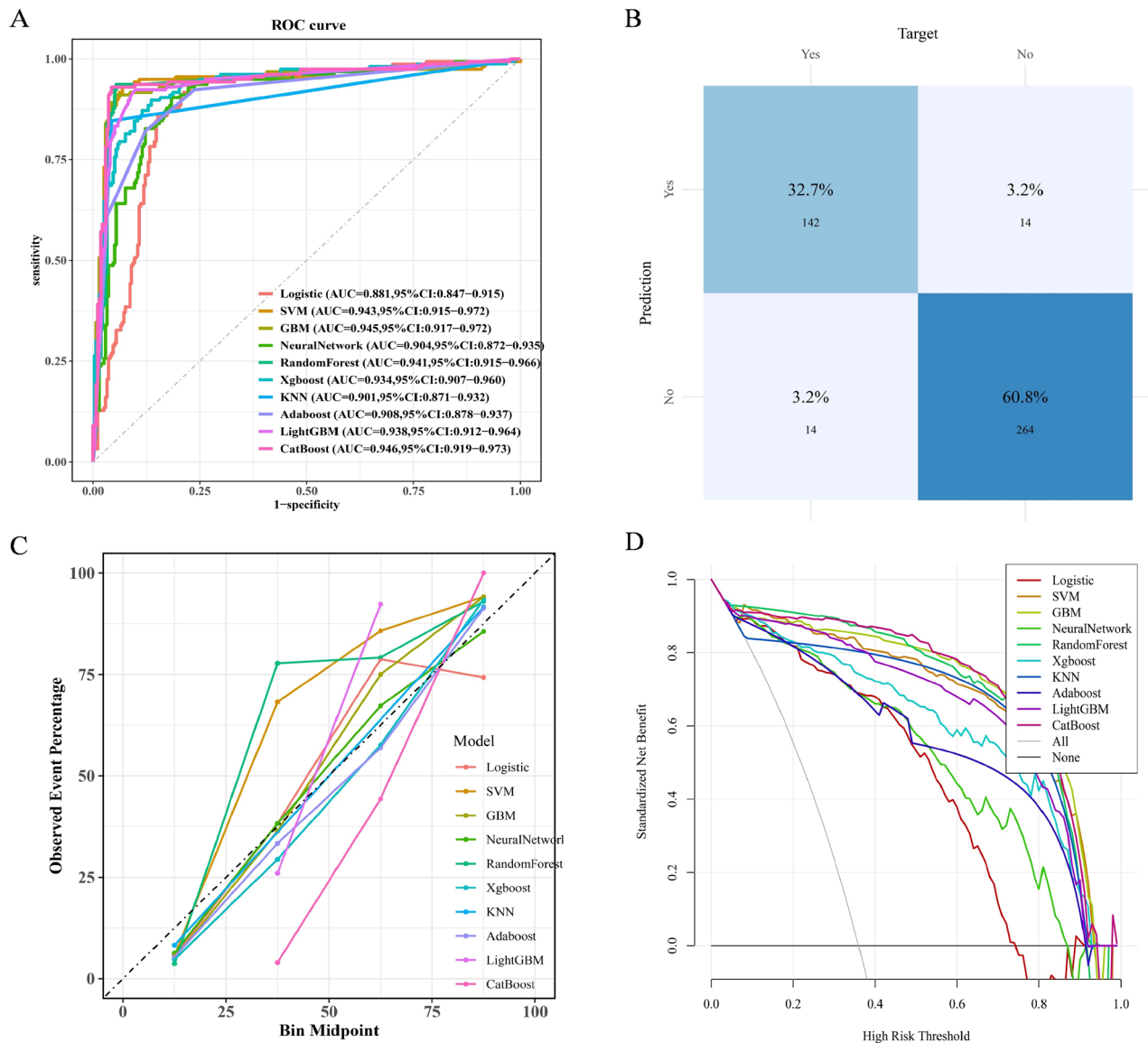


Fig. 2 Testing and evaluation of GBM model based on raw data. **(A)** AUC comparison between models. **(B)** The confusion matrix of GBM model. **(C)** Comparison of calibration curves between models. **(D)** Comparison of DCA curves between models

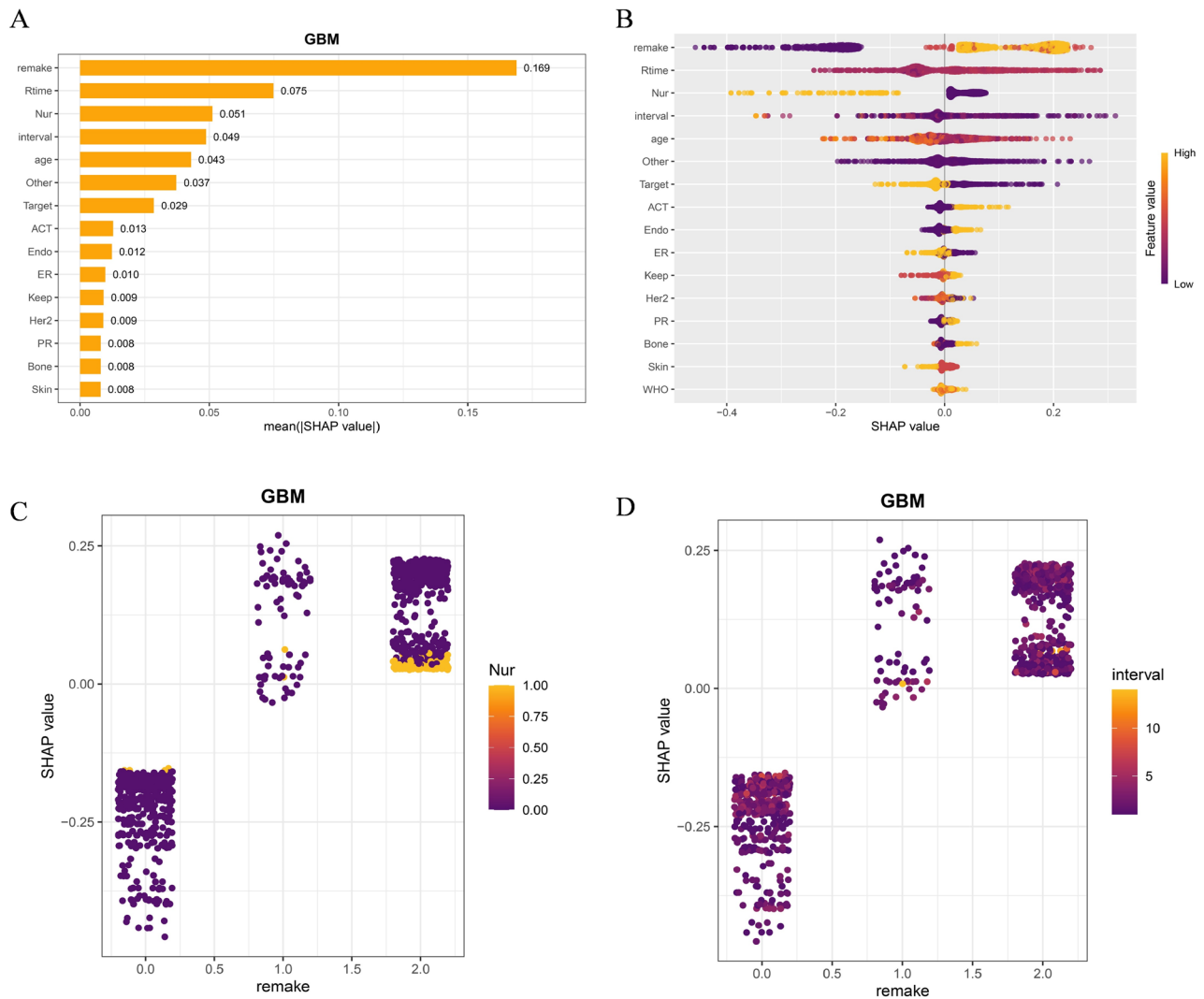


Fig. 3 Interpretation of the GBM model using SHAP. **(A)** Mean importance ranking of features displayed by SHAP. **(B)** Bee colony diagram of characterization attributes in SHAP. **(C)** Partial correlation dependence plot of remake ratio and perineural invasion ratio. **(D)** Partial correlation dependence plot of remake ratio and chemotherapy-to-surgery interval ratio

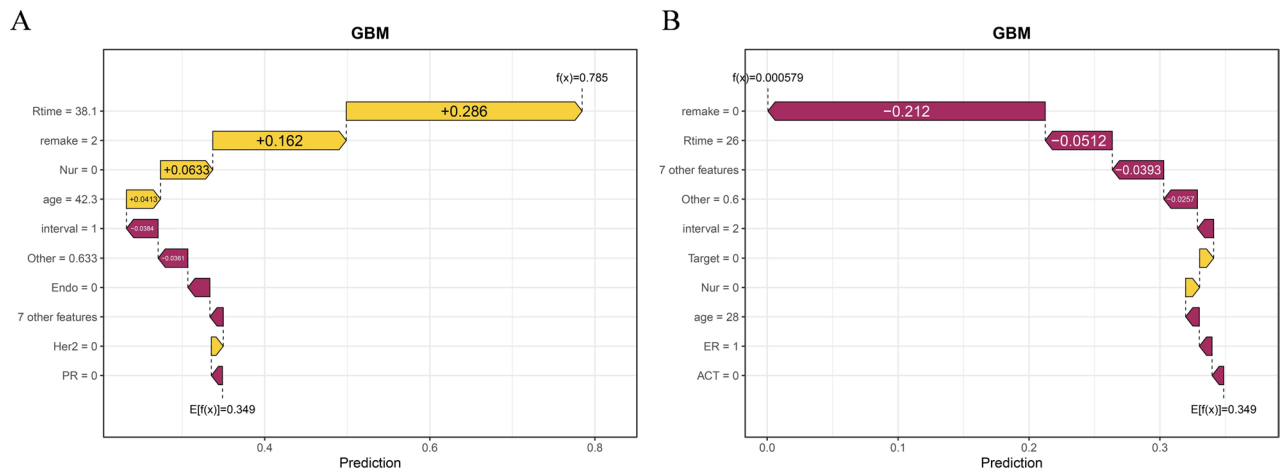


Fig. 4 Interpretation of the GBM model using SHAP. **(A)** A patient who underwent breast reconstruction and **(B)** a patient who did not undergo breast reconstruction

cost [25]. Our model, utilizing routinely collected clinicopathological data alone, offers a highly accurate, cost-effective, and readily implementable potential solution for personalizing PORT decisions in ypN0 patients.

SHAP analysis identified three paramount predictors—breast reconstruction, perineural invasion, and the chemotherapy-to-surgery interval—offering novel insights into the biological heterogeneity of radiotherapy response. The prominent role of breast reconstruction suggests a potential synergistic interaction with radiotherapy efficacy, extending beyond its restorative purpose. Plausible mechanisms include improved local tissue oxygenation via neovascularization in autologous flaps (which may enhance radiosensitivity) [26] and immune modulation triggered by surgical trauma and tissue transfer, potentially fostering an ‘in situ vaccination’ effect alongside radiotherapy [27]. However, the interpretation of this predictor requires caution. The decision to undergo reconstruction is associated with patient selection factors such as younger age, better performance status, and positive health attitudes—all potential confounders for better outcomes [28]. Therefore, while biologically plausible, this association likely reflects a combination of selection bias and potential biological effect, necessitating adjustment for these confounders in future prospective studies.

Consequently, for clinical interpretation, it is paramount to emphasize that breast reconstruction should not be interpreted as a modifiable oncologic risk factor or a target for intervention. Its prominence in our model is highly likely to be a proxy for multiple confounding factors associated with the selection for reconstruction, such as younger age, better performance status, fewer comorbidities, and positive health-seeking behavior. This finding underscores the importance of patient context in risk assessment but should not be used to guide decisions regarding reconstructive surgery.

The significance of perineural invasion reaffirms its established value as a marker of aggressive tumor biology. Its persistent negative impact even in ypN0 patients suggests the presence of underlying high-risk features, such as perineural micrometastases or an association with radioresistant molecular subtypes [29, 30]. This indicates a higher residual risk that may warrant consideration for treatment intensification or more vigilant follow-up, even in the absence of nodal disease.

Meanwhile, the chemotherapy-to-surgery interval likely captures inter-individual differences in physiological recovery from NAC toxicity within a standardized timeframe [31]. Our model effectively captures how variations in physiological reserve and recovery capacity influence a patient’s tolerance to and potential benefit from subsequent locoregional therapies. This highlights the clinical importance of moving beyond rigid

scheduling towards a more personalized assessment of a patient’s functional recovery, optimizing the sequence and impact of multimodal treatment.

Together, these predictors not only underpin the model’s accuracy but also appear to operate through largely independent biological and clinical pathways, as suggested by their distinct contributions in the SHAP analysis. This independence strengthens the model’s robustness and provides a transparent, multi-faceted foundation for personalizing adjuvant therapy decisions, positioning it as a tool that reflects the complex reality of clinical practice.

The application of SHAP transforms our model from a powerful yet opaque predictor into a transparent clinical reasoning aid [32]. By providing both global feature importance and local, individual-level explanations, the model empowers clinicians to understand the ‘why’ behind each prediction [33]. This fosters trust and facilitates the integration of data-driven insights with clinical expertise, moving beyond blanket recommendations towards truly tailored patient management.

This study has several limitations that must be considered when interpreting the results. First, the single-center, retrospective design introduces risks of selection bias and limits the generalizability of our findings; the model’s performance, while robust in internal validation, requires confirmation in independent, multi-center, prospective cohorts. Second, a key methodological limitation is that feature selection was performed on the entire dataset before splitting into training and test sets, which may lead to optimistic bias in the reported performance metrics (e.g., the high AUC). Consequently, the model’s discriminative ability should be interpreted with caution. Third, the absence of a control group of ypN0 patients not receiving PORT confines our model to a prognostic risk-stratification role within a treated population; it does not predict the absolute benefit of radiotherapy, and its value for personalizing PORT decisions remains hypothetical and requires testing in future randomized settings. Fourth, treating recurrence as a binary endpoint without formally accounting for differential follow-up time (Rtime) is a simplification; future studies employing time-to-event analysis frameworks (e.g., Cox models) are warranted. Fifth, although we addressed class imbalance through stratified sampling, its potential influence on precision metrics remains. Sixth, the prominence of features such as breast reconstruction in the SHAP analysis almost certainly reflects confounding by indication (e.g., association with younger age, better performance status) rather than a direct causal or modifiable risk factor, which must be clearly recognized to avoid clinical misinterpretation. Finally, the model does not incorporate potentially pivotal dynamic variables such as circulating tumor DNA

(ctDNA) or detailed genomic signatures, which represent opportunities for future enhancement.

In conclusion, this study developed and validated a highly accurate and interpretable machine learning model based on the GBM algorithm to stratify recurrence risk in ypN0 breast cancer patients “who call received” PORT. By leveraging the SHAP framework, the model provided transparent insights, identifying breast reconstruction, perineural invasion, and chemotherapy-to-surgery interval as key predictors. This tool, built from routine clinical data, represents a step towards personalized risk assessment within a specific patient cohort. Its potential clinical utility lies in identifying high-risk patients for intensified follow-up. However, it cannot currently guide decisions about whether to administer PORT, and its implementation requires prospective validation in external cohorts and in studies with appropriate control groups.

Abbreviations

NAC	Neoadjuvant chemotherapy
PORT	Postoperative radiotherapy
ML	Machine learning
SHAP	SHapley Additive exPlanations
PSO	Particle Swarm Optimization
GBM	Gradient Boosting Machine
AUC	Area under the receiver operating characteristic curve
DCA	Decision curve analysis
EMR	Electronic Medical Record
VIF	Variance inflation factor
LASSO	Least Absolute Shrinkage and Selection Operator
SVM	Support Vector Machine
RF	Random Forest
XGBoost	eXtreme Gradient Boosting
KNN	K-Nearest Neighbors
AdaBoost	Adaptive Boosting
LightGBM	Light Gradient Boosting Machine
XAI	Explainable Artificial Intelligence
pCR	Pathological complete response
LVI	Lymphovascular invasion
ER	Estrogen receptor
PR	Progesterone receptor
HER2	Human epidermal growth factor receptor 2

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Author contributions

Data analysis and initial draft preparation were performed by Yunjiao Zhang and Jie Duan. Panpan Liang, Xue Zhang, Xi Chen and Chutuo Liu contributed to study conception and provided critical revision of the manuscript. Yunjiao Zhang and Zhe Wang were responsible for data curation and validation. Rui Ling, Niuniu hou and Zhe Wang contributed to overall editing, supervision, and project administration. All authors have read and approved the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki and was approved by the Institutional Review Board of the First Affiliated Hospital of Air Force Medical University (No.: KY20232266-F-1). Due to the retrospective nature of the study and the use of de-identified data, the requirement for informed consent was waived by the ethics committee.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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